the wave function must instantaneously vanish, across the entire hemisphere, the moment that spot on the film indicates the electron hit. Anything less than instantaneous vanishing, and we run the risk of seeing a nonexistent second electron register on the film after the first, at some location where the wave function isn't yet zero. This "entirely peculiar mechanism of action at a distance," said Einstein, "implies to my mind a contradiction with the principle [i.e., the special theory] of relativity," which states very clearly that neither objects nor signals can travel faster than the speed of light. Thus, if quantum physics really is a complete description of nature, then it must violate relativity. To Einstein, the conclusion was obvious: the electron must have been in a particular location even before it hit the film, even though quantum physics could say nothing about where, exactly, it was. This was the only way Einstein saw to avoid invoking an instantaneous wave function collapse that would violate locality. Therefore, quantum physics was an incomplete description of nature, and more was needed to understand the true story of the quantum world. Specifically, to avoid contradictions with relativity, particles must have determinate locations at all times, in addition to wave functions. "I think that Mr. de Broglie is right to search in this direction," Einstein concluded.

The response to Einstein's thought experiment from the rest of the conference was muted incomprehension. Bohr, to his credit, admitted as much. "I feel myself in a very difficult position because I don't understand what precisely is the point which Einstein wants to [make]," he said. "No doubt it is my fault." Einstein's simple thought experiment offered a devastating critique of the Copenhagen position, but its simplicity may have paradoxically been a barrier to understanding: Einstein's presentation of it was quite brief and may have given the impression that he was confused about the nature of probability. Bohr, in particular, seems to have misunderstood Einstein rather badly: he recalled later that Einstein had doubts about Heisenberg's uncertainty principle, and introduced his thought experiment because he was looking for ways around it. Einstein's concerns about locality went unheard for the rest of the Solvay conference. But Einstein soon developed new thought experiments, doggedly pursuing the problems he saw with quantum physics.